PROBABILITY LADDER 2 OF 9

Counting: arrangements and choices

Probabilities of equally likely outcomes need counts, and the multiplication principle and binomial coefficients supply them.

1 / THE RULE AND THE PICTURE

Choosing 2 of 4 in order gives 12 pairs, and ignoring order halves each into 6

first \ second	A	B	C	D
A		AB	AC	AD
B	BA		BC	BD
C	CA	CB		CD
D	DA	DB	DC	

first pick down the side, with the gold pairs counted once

There are 4 choices for the first pick and 3 for the second, so $4 \cdot 3 = 12$ ordered pairs (AB and BA differ). If order does not matter each pair is counted $2!$ times, so the number of unordered pairs is $12/2 = 6$:

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!} = \frac{n(n-1)\cdots(n-r+1)}{r!}$$

2 / ARRANGEMENTS / THE MULTIPLICATION PRINCIPLE

Independent choices multiply, and a repeated letter divides the count

situation	count	value
5 books on a shelf	$5!$	120
podium of 3 from 10 runners	$10 \cdot 9 \cdot 8$	720
plate: 3 letters, 2 digits	$26^3 \cdot 10^2$	$1\,757\,600$
arrangements of MISSISSIPPI	$\frac{11!}{1!\,4!\,4!\,2!}$	$34\,650$

If a task has a ways for the first stage and b for the second, there are ab ways in all. Swapping identical letters changes nothing, so MISSISSIPPI divides $11!$ by $4!$ for the I's, $4!$ for the S's and $2!$ for the P's.

3 / CHOICES / ORDER DOES NOT MATTER

Binomial coefficients count the ways to pick a group

situation	count	value
handshakes among 10 people	$\binom{10}{2}$	45
committee of 3 from 8	$\binom{8}{3}$	56
five-card poker hands	$\binom{52}{5}$	$2\,598\,960$
lottery, 6 from 49	$\binom{49}{6}$	$13\,983\,816$

The count is symmetric, $\binom{n}{r} = \binom{n}{n-r}$, since picking r people to include is the same as picking $n-r$ to leave out, and $\binom{49}{6} = 13\,983\,816$ is why the lottery is hard to win.

4 / PASCAL'S TRIANGLE / THE TABLE OF COEFFICIENTS

Each coefficient is the sum of the two above it, and each row sums to 2^n

$n=0$	1	$2^0 = 1$
$n=1$	1 1	$2^1 = 2$
$n=2$	1 2 1	$2^2 = 4$
$n=3$	1 3 3 1	$2^3 = 8$
$n=4$	1 4 6 4 1	$2^4 = 16$
$n=5$	1 5 10 10 5 1	$2^5 = 32$
$n=6$	1 6 15 20 15 6 1	$2^6 = 64$

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}, \qquad \binom{6}{2} = \binom{5}{1} + \binom{5}{2} = 5 + 10 = 15.$$

Picking a person to be in or out gives two choices each, so the 2^n subsets of n people split by size as the row $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$, and the sum of the row is 2^n. The same coefficients expand $(a+b)^n$.

5 / FROM COUNTS TO PROBABILITIES

Probability is favourable choices over all choices

For a poker hand the full house has $13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2} = 3744$ choices out of $\binom{52}{5} = 2\,598\,960$, a probability of 0.144%.

Two cards drawn from a pack are both aces with probability $\binom{4}{2}/\binom{52}{2} = \frac{6}{1326} = \frac{1}{221}$, and a single lottery ticket wins with probability $1/13\,983\,816$.

ASIDE / ORDER MATTERS WHEN

Counting arrangements uses $n!$ and counting groups uses $inomnr$, and the only difference is dividing out the $r!$ orders.

LINKS TO OTHER SHEETS

The card-pack entropy sheet uses the same poker counts; conditional draws are in the next sheet, *Conditional probability and independence*.

Convention. $n! = n(n-1)\cdots 1$; $\binom{n}{r}$ counts unordered choices; the cards and people are distinct unless stated.

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